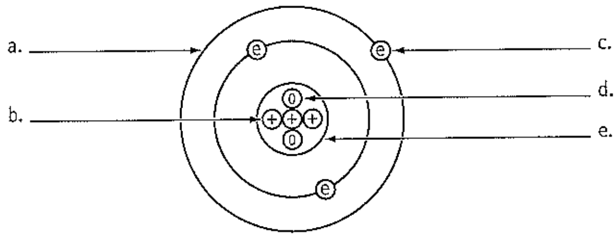
Atomic structure

Note: You will need to refer to a copy of the periodic table for this worksheet.

A popular atomic model

Many people today believe that atoms can be described by the atomic model of the solar system atom. This orbital atomic model allows us to explain many processes.

The atom is made up of three main subatomic particles which are grouped into two lots. Protons and neutrons are tightly packed into the nucleus which forms the centre of the atom. Electrons surround the nucleus on several orbits (or shells).



1 Label the diagram of an atomic model (lithium).

Subatomic particles

Protons and neutrons are 2000 times heavier than electrons. Protons and electrons have an electric charge: protons are positive and electrons are negative. Electrons fly around the nucleus on orbits. Neutrons and protons rest in the nucleus.

2 Complete the table about subatomic particles.

Name	Charge	Mass	Position	Movement
			on orbits	
		2000 times		
	neutral			no

Atomic numbers and mass numbers, atoms and isotopes

Atoms are neutral, although some of their subatomic particles do have an electric charge. But the positive and negative subatomic charges cancel each other leaving the atom neutral. Therefore, the number of positive protons must equal the number of negative electrons. The *atomic number* gives the number of protons.

Since protons and neutrons are very much heavier than electrons, it is the protons and neutrons which determine the mass of the atoms. Usually an atom has as many neutrons as protons. The *mass number* is the sum of protons and neutrons.

Some elements have atoms which have the same number of protons in the nucleus (so the same atomic number) but more or fewer neutrons (so they have different atomic weights). Their atoms are now called *isotopes*.

When the number of neutrons is much larger than the number of protons in a large nucleus, the isotope's nucleus is unstable. As a result it breaks up, forming new atoms and giving off radiation.

3 Complete the table about mass and atomic numbers of atoms.

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	${}^{5}_{3}Li$
magnesium	24				12	
aluminium		13		13		
hydrogen	1					
chlorine						${}^{35}_{17}Cl$
phosphorus						${}^{31}_{15}P$
oxygen						${}^{16}_{8}O$

Atomic structure

Electrons on shells

Electrons move on orbits (or shells) around the nucleus. The inside shells are smaller than the outer shells and so can hold fewer electrons. Inner shells are filled up before the outer shells. The first innermost shell can carry a maximum of two electrons, the second and third shells can hold no more than eight electrons (these maximum numbers are true for the first 20 atoms).

4 Draw atomic models for the following atoms. The mass and atomic numbers give you a clue to the number of protons, neutrons and electrons.

helium ($^{4}_{2}He$)

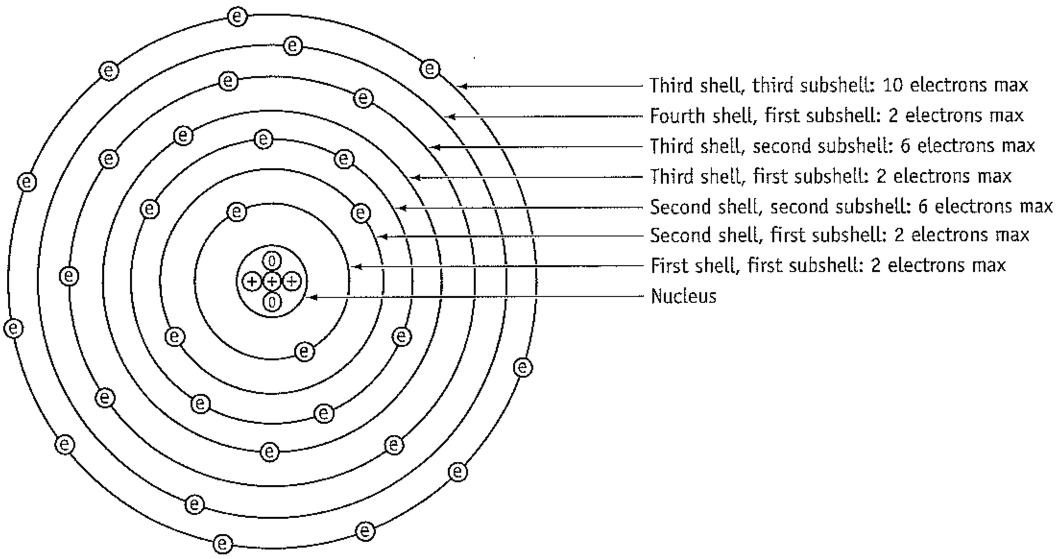
silicon ($^{28}_{14}Si$)

beryllium ($^{9}_{4}Be$)

potassium ($^{39}_{19}K$)

Shells and their subshells

The orbital model of the atom becomes untidy with the twenty-first atom, because the electron shells are actually composed of up to four subshells. And some of the subshells overlap each other. The third subshell of the third shell is actually bigger than the first subshell of the fourth shell.



5 State the number of protons, neutrons and electrons for the atom of iron ($^{56}_{26}Fe$).

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6 Shade in electrons to the atomic model above for iron.

7 State the number of subshells for the first three shells. ____________________________________

The periodic table of elements

A brief history

In 1829 scientists knew about 40 elements. German scientist Johann Dobereiner noticed that lithium, sodium and potassium had very similar properties. Calcium, strontium and barium were also similar to each other. Chlorine, bromine and iodine formed another triad. In 1864, English scientist John Newlands replaced the triads with a law of octaves, which was based on the atomic weights of 60 known elements as well as their properties. Dmitri Mendeleev, from Russia, refined the periodic table in 1869, leaving gaps for some elements which had not been discovered. Mendeleev predicted the properties of some of those missing elements quite accurately. However, there were some inconsistencies: certain elements were too heavy for their position in the periodic table. Henry Mosely (England, 1914) resolved these problems, when he based the order of the elements on their atomic number rather than their weight. Today we know 103 elements and they are all organised into the periodic table of elements.

Some properties of the first 20 elements

Name	Symbol	Atomic weight	Order by weight	Atomic number	Atomic radius (pm)	Melting point (°C)
Aluminium	Al	27		13	143	660
Argon	Ar	40		18	94	– 189
Beryllium	Be	9		4	112	1280
Boron	B	11		5	88	2030
Calcium	Ca	40		20	197	838
Carbon	C	12		6	77	3730
Chlorine	Cl	35		17	99	–101
Fluorine	F	19		9	64	–220
Helium	He	4		2	50	–270
Hydrogen	H	1		1	37	–259
Lithium	Li	7		3	152	1650
Magnesium	Mg	24		12	160	650
Neon	Ne	20		10	60	–249
Nitrogen	N	14		7	70	–210
Oxygen	O	16		8	66	–219
Phosphorus	P	31		15	110	44
Potassium	K	39		19	231	64
Silicon	Si	28		14	118	1410
Sodium	Na	23		11	186	98
Sulfur	S	32		16	102	113

Note: pm = picometre = 10^{-12}m = 1/ 1 000 000 000 000 metres

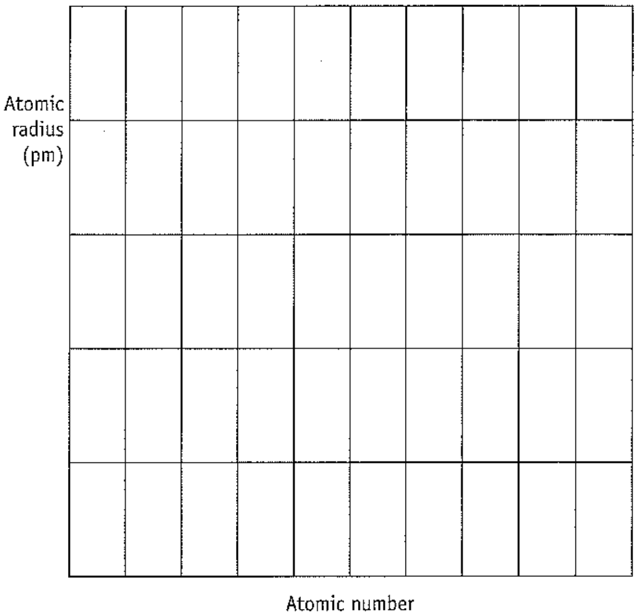
1 Order the elements by weight and write their order numbers into the appropriate column above.

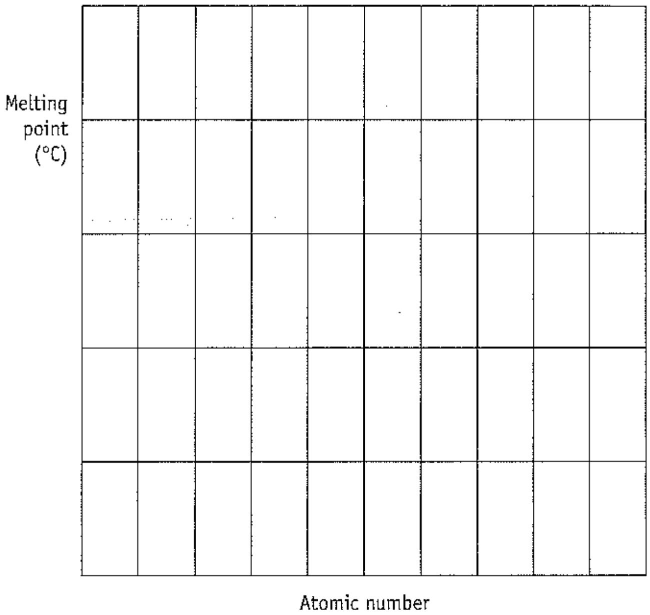
2 State at which point (by element name) this order is different to the atomic number order.

3 Graph the atomic radius on the vertical axis against the atomic number on the horizontal axis on the first grid on the next page.

4 Graph the melting point against the atomic number on the second grid.

The periodic table of elements





The periodic table for the first 20 elements

5 Complete the periodic table for the first 20 elements below by writing the symbols into the spaces. Let the atoms with a large atomic radius begin the horizontal periods. Let the elements with a low melting point finish the periods. Let the other elements fit into the periodic table in order of their size and melting point. The first element, hydrogen, is often written above the periodic table, because it has properties from the first and the seventh horizontal group.

						1	2
3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18
19	20						

Trends in the periodic table

Form a diagonal line across the periodic table by shading in the cells of boron and silicon. To the left of this line are the metals, to the right of the line are the non-metals. The shaded elements are semi-metals.

6 Complete the table below by summarising some general properties of metals and non-metals.

Aspect	Metals	Non-metals
Conduct electricity?		
Conduct heat?		
Are solid?		
Are malleable?		

7 Complete the table below by answering 'yes' or 'no' about changes in size, weight and properties of atoms.

Property	Within periods (going across)	Within groups (going down)
Atomic radius increases		
Atomic weight increases		
Atoms have similar properties		